CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

12 JANUARY 2017 (p.m.)

J1701207020

FILL IN ALL THE INFORMATION REQUESTED CLEARLY IN CAPITAL LETTERS.

TEST CODE | 0 | 1 | 2 | 0 | 7 | 0 | 2 | 0 |

SUBJECT BIOLOGY – Paper 02

PROFICIENCY GENERAL

REGISTRATION NUMBER

SCHOOL/CENTRE NUMBER

NAME OF SCHOOL/CENTRE

CANDIDATE'S FULL NAME (FIRST, MIDLE, LAST)

DATE OF BIRTH | D | D | M | M | Y | Y | Y | Y |

SIGNATURE _____

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FORM TP 2017002

JANUARY 2017

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®

EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, answers should be illustrated by diagrams.
5. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
6. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) A group of students conducts an experiment to investigate the rate of reaction of Enzyme X at different temperatures. The results of the experiment are shown in Table 1.

TABLE 1: RATE OF REACTION OF ENZYME X AT DIFFERENT TEMPERATURES

Temperature (°C)	Rate of Reaction of Enzyme X (mg of product per min)
0	0
5	3
15	9
25	15
30	24
40	8
50	0

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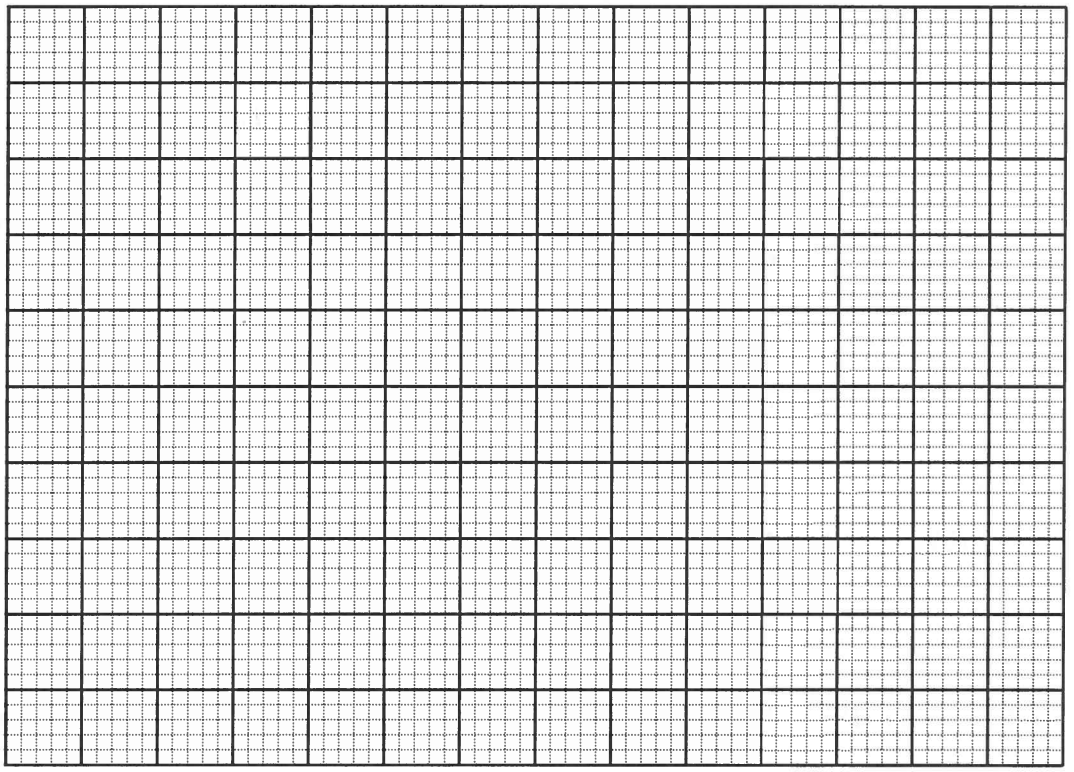
(i) Plot the data in Table 1 on the grid below.



(4 marks)

(ii) Use the graph plotted in (a) (i) to identify the optimum temperature for Enzyme X.

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(1 mark)

(iii) Account for the shape of the graph:

Between 5 °C and 30 °C

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(2 marks)

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Above 40 °C

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(2 marks)

(iv) State ONE precaution that the students should take when carrying out this experiment.

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(1 mark)

(b) (i) Name TWO enzymes that break down protein.

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(2 marks)

(ii) Name TWO parts of the human alimentary canal where enzymes that break down protein are found.

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(2 marks)

(iii) Suggest TWO reasons why the different enzymes for breaking down proteins are located in two different parts of the alimentary canal.

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(2 marks)

(c) Leguminous plants such as peanuts are able to manufacture proteins with the help of nitrogen-fixing bacteria that live in the nodules on their roots.

(i) Identify the type of relationship that exists between the peanut plant and the bacteria.

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(1 mark)

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(ii) Explain your answer in (c) (i) on page 6.

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(3 marks)

(iii) Describe a suitable procedure for testing the presence of protein in peanuts.

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(3 marks)

(iv) Suggest TWO reasons why plants need protein.

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(2 marks)

Total 25 marks

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2. (a) (i) State THREE differences between a plant cell and an animal cell.

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(3 marks)

(ii) State TWO differences between a prokaryotic cell and a eukaryotic cell.

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(2 marks)

(b) State the function of EACH of the following:

(i) Cell wall

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(1 mark)

(ii) Cell membrane

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(1 mark)

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(c) (i) Explain what happens to an animal cell when it is removed from the organism and placed into concentrated sucrose solution.

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(2 marks)

(ii) Explain what happens to an animal cell when it is removed from the organism and placed into distilled water.

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(2 marks)

(iii) Describe how the results would be different from (c) (i) and (ii) for a plant cell in concentrated sucrose solution and distilled water respectively.

Concentrated sucrose solution

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(2 marks)

Distilled water

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(2 marks)

Total 15 marks

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3. (a) Figure 1 shows components of blood.

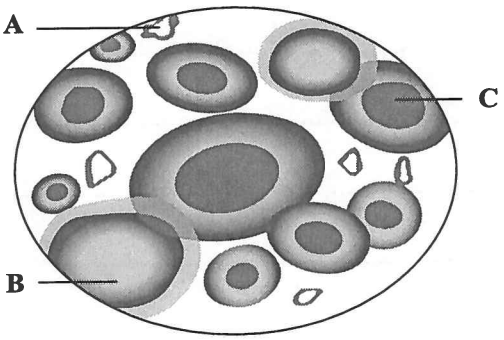

Figure 1. Components of blood

(i) Identify the components of blood that are labelled A, B and C in Figure 1.

A .....

B .....

C .....

(3 marks)

(ii) Name the component of the blood that is responsible for clotting.

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(1 mark)

(b) Complete the flow diagram shown in Figure 2 by filling in the missing words at positions D, E and F. **(3 marks)**

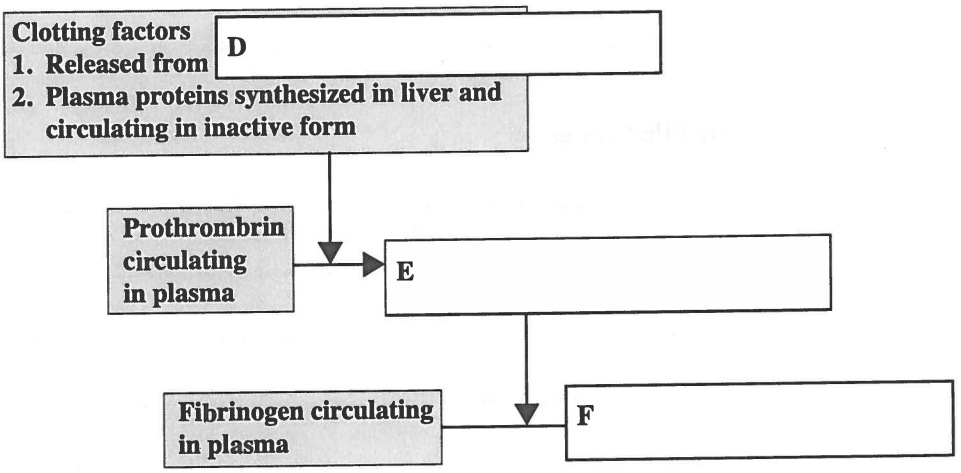

Figure 2. Diagram showing steps in blood clotting

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(c) Sometimes persons may inherit a condition called haemophilia in which the blood does not clot normally. The allele for blood clotting is located on the X chromosome. A female with normal blood clotting ability mates with a male with normal blood clotting ability to produce offspring

(i) Complete the following genetics diagram to illustrate this information.

Parents' phenotypes: Male × Female

Parents' genotypes: ..... × X^HX^h

Gametes: ..... × $(X^H)(X^h)$

Fertilization cross:

Gametes ♀ ♂	X^H	X^h
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(3 marks)

(ii) State the probability of the couple having a haemophiliac child.

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(1 mark)

(iii) What is the likely sex of the haemophiliac child? Give ONE reason to support your answer.

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(3 marks)

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(d) Suggest ONE reason why haemophiliacs may be more susceptible to infectious disease.

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(1 mark)

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

4. (a) Describe how oxygen in the atmosphere becomes available to human blood for delivery to the cells, and explain how this process differs from the way plant cells obtain oxygen for respiration.

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(9 marks)

(b) Suggest SIX reasons why humans who smoke heavily have less oxygen available to the cells for respiration.

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(6 marks)

Total 15 marks

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5. (a) (i) Blood that has been oxygenated in the lungs must first be pumped through the heart before it is sent to all the other organs of the body. Describe the pathway that blood takes as it flows into the heart from the lungs, until it is pumped out of the heart to be sent to the rest of the body.

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(4 marks)

(ii) Identify TWO physiological diseases that affect the circulatory system.

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(2 marks)

(b) Chemotherapy is a treatment that destroys both the malignant (bad) cells and the good cells in the body. Explain why a person undergoing chemotherapy would have decreased natural immunity.

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(4 marks)

(c) Peter, who has not had a tetanus vaccine, gets his foot punctured by a nail. He is given a tetanus antiserum injection at the hospital. Suggest why he is given an antiserum injection instead of a vaccine.

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(5 marks)

Total 15 marks

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6. (a) Describe how genetic engineering is used to produce human insulin using the bacterium *E. coli*.

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(6 marks)

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(b) (i) Discuss THREE differences between genetic engineering and artificial selection used by farmers to produce improved varieties of crop plants.

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(6 marks)

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(ii) Suggest THREE advantages of using genetic engineering to produce insulin.

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(3 marks)

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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EXTRA SPACE

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Question No. ☐

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EXTRA SPACE

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CANDIDATE'S RECEIPT

INSTRUCTIONS TO CANDIDATE:

1. **Fill in all the information requested clearly in capital letters.**

TEST CODE: | 0 | 1 | 2 | 0 | 7 | 0 | 2 | 0 |

SUBJECT: BIOLOGY – Paper 02

PROFICIENCY: GENERAL

REGISTRATION NUMBER: | | | | | | | | | | |

FULL NAME: _____
(BLOCK LETTERS)

Signature: _____

Date: _____

2. **Ensure that this slip is detached by the Supervisor or Invigilator and given to you when you hand in this booklet.**

3. **Keep it in a safe place until you have received your results.**

INSTRUCTION TO SUPERVISOR/INVIGILATOR:

Sign the declaration below, detach this slip and hand it to the candidate as his/her receipt for this booklet collected by you.

I hereby acknowledge receipt of the candidate's booklet for the examination stated above.

Signature: _____
Supervisor/Invigilator

Date: _____

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